UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
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Lung Cancer CT-Scan Classification using Transfer Learning-Based CNN Architectures and Vision Transformer

Course: Machine Learning Theory

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Abstract

Lung cancer is one of the leading causes of cancer-related mortality worldwide, and early, reliable diagnosis from computed tomography (CT) scans plays a critical role in improving patient outcomes. This project presents a comparative deep learning framework for the classification of lung CT-scan images into three categories: Benign, Malignant, and Normal, using the publicly available IQ-OTH/NCCD Lung Cancer dataset. Six transfer-learning-based Convolutional Neural Network (CNN) architectures – ResNet50, ResNet152, EfficientNetB0, VGG16, InceptionV3, and MobileNetV2 – pretrained on ImageNet, are fine-tuned with a common lightweight classification head, and their performance is compared against a Vision Transformer (ViT) trained from scratch. All models are trained under identical preprocessing, augmentation, and training conditions for 30 epochs to ensure a fair benchmark. Experimental results show that VGG16 achieved the highest test accuracy of 86.67%, followed by EfficientNetB0 (84.00%) and ResNet152 (82.50%), while the Vision Transformer, trained without pretraining, achieved only 45.00%, confirming that transformer architectures require substantially more data to compete with pretrained CNNs on small medical imaging datasets. Detailed classification reports, confusion matrices, and comparative visualizations are presented to support these findings.

1 Introduction

Lung cancer remains one of the most fatal forms of cancer worldwide, accounting for the highest number of cancer-related deaths among both men and women across the globe. According to global cancer statistics, lung cancer is diagnosed in over two million new patients every year, and its five-year survival rate remains alarmingly low compared to many other cancer types. The primary reason behind this poor prognosis is that lung cancer is frequently diagnosed at an advanced stage, when the tumor has already grown large or metastasized to other organs, leaving limited and often less effective treatment options. Early detection, by contrast, is strongly associated with significantly improved survival outcomes, which makes the development of accurate, fast, and reliable diagnostic tools a critical priority in modern oncology and radiology.

Computed Tomography (CT) imaging is the primary non-invasive tool used by radiologists to detect and characterize pulmonary nodules, owing to its ability to produce high-resolution, cross-sectional images of the lungs that reveal the size, shape, density, and location of suspicious growths. However, manual interpretation of CT scans is time-consuming, requires years of specialized radiological expertise, and is subject to inter-observer variability, meaning that two equally qualified radiologists may arrive at different conclusions when examining the same scan. Subtle nodules can be easily missed on a single reading, especially in busy clinical settings where radiologists must review a large volume of scans under time pressure. These limitations create a strong motivation for computer-aided diagnosis (CAD) systems that can assist radiologists by flagging suspicious regions, providing a second opinion, or automating routine triage decisions.

Deep learning, and particularly Convolutional Neural Networks (CNNs), has emerged as a powerful tool to automate and support this diagnostic process by learning discriminative visual patterns directly from imaging data, without requiring hand-crafted feature engineering. Over the past decade, CNN architectures pretrained on large-scale natural image datasets such as ImageNet have been successfully repurposed for medical imaging tasks through transfer learning, allowing high-performing models to be trained even when only a modest amount of labeled medical data is available. More recently, Vision Transformers (ViTs) have challenged the dominance of CNNs in general computer vision by relying entirely on self-attention mechanisms rather than convolutional

filters, raising the natural question of whether transformer-based architectures can match or exceed CNN performance in specialized, data-scarce domains such as lung CT-scan classification.

This project investigates the use of deep learning models to automatically classify lung CT-scan images into three diagnostic categories – Benign, Malignant, and Normal – using the publicly available IQ-OTH/NCCD Lung Cancer dataset [1]. Rather than relying on a single architecture, as is common in much of the existing literature, this study systematically compares six widely-used transfer-learning-based CNN backbones – ResNet50, ResNet152, EfficientNetB0, VGG16, InceptionV3, and MobileNetV2 – against a Vision Transformer trained entirely from scratch, under one unified experimental pipeline. All seven models are trained with identical preprocessing, augmentation, optimizer, loss function, and epoch settings, so that any observed difference in performance can be attributed to the architecture itself rather than to inconsistencies in the experimental setup. This design choice distinguishes the present work from many published studies, which often tune each architecture separately and therefore make it difficult to draw fair conclusions about which backbone is genuinely best suited to the task.

The remainder of this report is organized as follows. Section 2 reviews the background concepts underlying CNNs, transfer learning, and Vision Transformers that are relevant to this study. Section 3 describes the proposed methodology, including the dataset, preprocessing pipeline, loss function, the six CNN architectures, the from-scratch Vision Transformer, and the shared classification head used to ensure a fair comparison. Section 4 presents the experimental results, including accuracy comparisons, per-class classification reports, confusion matrices, and a comparison with state-of-the-art literature on the same dataset. Finally, Section 5 concludes the report and outlines directions for future work.

1.1 Research Challenges

Several challenges motivate and complicate this study:

- **Limited dataset size.** Medical imaging datasets, including IQ-OTH/NCCD, are considerably smaller than large-scale natural image datasets such as ImageNet, which makes training deep models from scratch prone to overfitting.
- **Visual similarity between classes.** Benign and malignant lesions can appear visually similar on CT slices, particularly in early stages, making fine-grained discrimination difficult even for pretrained networks.
- **Choice of architecture.** A wide variety of CNN families exist (residual networks, lightweight mobile networks, multi-scale inception networks, compound-scaled networks), each with different inductive biases, and it is not obvious a priori which is best suited to CT-scan texture patterns.
- **Data-hungry transformer architectures.** Vision Transformers lack the built-in inductive biases (locality, translation invariance) of CNNs and typically require large-scale pretraining to perform competitively, which is not readily available for specialized medical domains.
- **Fair and reproducible comparison.** Comparing architectures fairly requires that all models be trained with identical image sizes, augmentation strategies, optimizers, loss functions, and number of epochs – a methodological challenge that many existing studies do not fully control for.

1.2 Novel Contributions

The main contributions of this project are summarized as follows:

- A unified and reproducible benchmarking pipeline is designed in which six pretrained CNN backbones (ResNet50, ResNet152, EfficientNetB0, VGG16, InceptionV3, MobileNetV2) and a Vision Transformer are all trained under identical preprocessing, augmentation, optimizer, loss function, batch size, and epoch settings, enabling a fair, apples-to-apples comparison.
- A common lightweight classification head (Global Average Pooling $\rightarrow$ Dense(128, ReLU) $\rightarrow$ Dropout(0.3) $\rightarrow$ Dense(3, Softmax)) is attached uniformly to every pretrained backbone, isolating the effect of the backbone architecture from the effect of the classifier design.
- A Vision Transformer is implemented from scratch (patch embedding, learnable position embedding, multi-head self-attention, and MLP blocks) without any pretrained weights, to empirically study how a transformer-based model behaves relative to pretrained CNNs on a small, three-class medical imaging dataset.
- An extensive comparative evaluation is carried out using accuracy, per-class precision, recall, F1-score, and confusion matrices, along with a consolidated bar-chart comparison of all seven models, providing practical guidance on model selection for lung CT-scan classification tasks.

2 Background Knowledge Based on Deep Learning Methods

Deep learning has become the dominant paradigm for medical image analysis due to its ability to automatically learn hierarchical feature representations directly from raw pixel data, removing the need for hand-crafted feature engineering that characterized earlier machine learning approaches.

Convolutional Neural Networks (CNNs). CNNs use learnable convolutional filters to detect local visual patterns (edges, textures, shapes) in early layers, which are progressively combined into more abstract, task-relevant representations in deeper layers. This locality and weight-sharing gives CNNs a strong inductive bias for image data and allows them to generalize well even with moderately sized training sets [2, 3].

Transfer Learning. Training a deep CNN from scratch requires large amounts of labeled data, which is rarely available in medical imaging. Transfer learning addresses this by taking a network pretrained on a large, generic dataset (typically ImageNet, containing over a million natural images across 1000 classes) and reusing its learned convolutional filters as a fixed or fine-tunable feature extractor for a new, related task. Because low- and mid-level visual features (edges, textures, blobs) are largely transferable across domains, transfer learning drastically reduces the amount of labeled data and training time needed to reach good performance on a new dataset such as lung CT scans [1].

Residual Learning. Very deep networks suffer from vanishing gradients and degradation problems during training. Residual networks (ResNet) solve this by introducing skip (identity) connections that allow gradients to flow directly across layers, enabling networks with over 100 layers to be trained effectively [2].

Compound Scaling. EfficientNet architectures scale network depth, width, and input resolution together in a principled way, achieving strong accuracy with fewer parameters and lower computational cost than earlier hand-designed CNNs [4].

Multi-Scale Feature Extraction. Inception-style networks (e.g., InceptionV3) apply convolutional filters of multiple sizes (1×1, 3×3, 5×5) in parallel within the same layer, allowing the network to simultaneously capture fine and coarse spatial patterns [5].

Lightweight Mobile Architectures. MobileNet-family networks use depthwise-separable convolutions to drastically reduce the number of parameters and floating-point operations, making them suitable for resource-constrained or real-time deployment scenarios, often at a modest accuracy trade-off [6].

Vision Transformers (ViT). Originally developed for natural language processing, the Transformer architecture relies on the self-attention mechanism to model relationships between all elements of a sequence simultaneously. The Vision Transformer adapts this idea to images by splitting an image into fixed-size patches, linearly embedding each patch, adding learnable positional embeddings, and processing the resulting sequence of patch embeddings through stacked self-attention and feed-forward blocks. Unlike CNNs, ViTs have no built-in notion of locality or translation invariance and must learn these properties from data, which is why they typically require very large pretraining datasets (or transfer learning from such pretraining) to match or exceed CNN performance [7].

3 Proposed Methodology

This section describes the dataset, preprocessing pipeline, loss function, model architectures, the proposed classification framework, and the evaluation metrics used in this study.

3.1 Dataset

The IQ-OTH/NCCD Lung Cancer dataset is used, which contains CT-scan images of the chest organized into three diagnostic classes: Benign cases, Malignant cases, and Normal cases. The dataset used in this project was already partitioned into training, validation, and test splits, as summarized in Table 1.

Table 1: Dataset split summary

Split	No. of Training	No. of Validation	Testing
Training	3000	3	1000
Validation	600	3	200
Test	600	3	200

Since all three classes are equally represented in both the validation and test sets (200 images per class), the dataset is balanced, which is an important consideration for the choice of loss function discussed in Section 3.3.

3.2 Image Preprocessing Methods

A uniform preprocessing pipeline is applied to all models. All CT-scan images are first resized to 128×128 pixels using `image_dataset_from_directory`, which keeps computation manageable while retaining sufficient visual detail, and are then grouped into mini-batches of size 32 for efficient GPU training. The `tf.data` pipeline additionally uses `.cache()` and `.prefetch(AUTOTUNE)` to overlap data loading with model computation and speed up training. Because each pretrained backbone expects pixel values scaled in a particular way, every model uses its own dedicated `preprocess_input` function (e.g., `resnet.preprocess_input`, `efficientnet.preprocess_input`, `vgg16.preprocess_input`, `inception_v3.preprocess_input`, `mobilenet_v2.preprocess_input`), ensuring that pixel values are scaled exactly as expected by the corresponding ImageNet-pretrained

weights. Finally, to artificially increase training diversity and reduce overfitting on the relatively small dataset, data augmentation – consisting of a random horizontal flip, random rotation (±10%), and random zoom (±10%) – is applied on-the-fly using Keras preprocessing layers during training only.

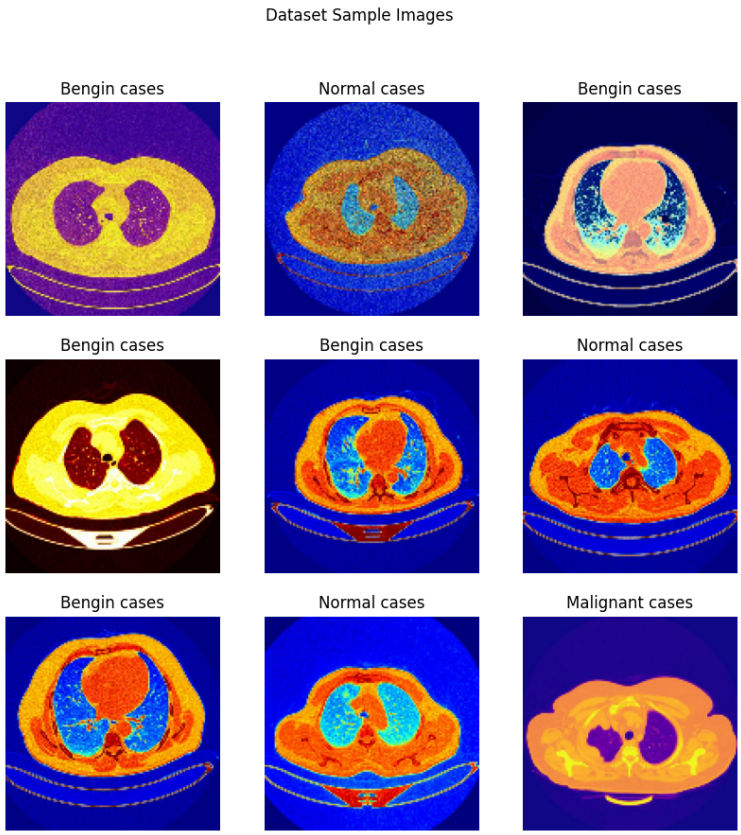


Figure 1: Representative sample CT-scan images from the training set, one grid covering all three classes (Benign, Malignant, Normal).

3.3 Loss Function

A central design decision in any classification pipeline is the choice of loss function. Focal Loss is a well-known alternative to standard cross-entropy that was originally proposed for object detection, where it down-weights the loss contribution of easy, well-classified examples and focuses training on hard, misclassified examples – making it particularly useful for datasets with severe class imbalance [8]. It is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \tag{1}$$

where p_t is the model's estimated probability for the true class, γ is a focusing parameter that reduces the loss for easy examples, and α_t is a class-balancing weight.

In this project, however, the training, validation, and test sets are all perfectly balanced across the three classes (Section 3.1). Since class imbalance is the primary motivation for using Focal Loss, and no such imbalance is present in this dataset, the sparse categorical cross-entropy loss was

used instead:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{2}$$

where $C = 3$ is the number of classes, y_i is the true label (one-hot encoded internally), and $\hat{y}_i$ is the predicted softmax probability for class i. All seven models in this study are optimized using the Adam optimizer with this loss function, ensuring a controlled and fair comparison. Focal Loss remains a promising direction for future work if the dataset were extended with a more imbalanced or larger-scale patient cohort.

3.4 Transfer Learning-Based CNN Architectures

Six ImageNet-pretrained CNN backbones are used as feature extractors. In every case, the convolutional base is frozen (its ImageNet weights are not updated) and only the new classification head (Section 3.6) is trained, which is fast and effective on a dataset of this size.

3.4.1 VGG16

VGG16 [3] is one of the earliest and most straightforward deep CNN architectures, built entirely from stacks of small 3×3 convolutional filters followed by max-pooling layers, with 16 weight layers in total. Its simple, uniform design makes it an excellent classical baseline against which more modern architectures can be compared. The use of small, stacked 3×3 filters rather than a few large filters gives VGG16 a large effective receptive field while keeping the number of parameters per layer relatively low, and it also lets the network apply more non-linear activations for the same receptive field, which improves its discriminative power. Because every convolutional block follows the same simple pattern, VGG16 is easy to fine-tune and its ImageNet-pretrained filters, which primarily capture edges, textures, and local contrast, transfer particularly well to grayscale-like, texture-dominated medical images such as chest CT slices. In this project, VGG16 was frozen and used purely as a feature extractor beneath the shared classification head. Despite being the oldest and structurally simplest architecture evaluated in this study, VGG16 achieved the highest test accuracy of all seven models, suggesting that architectural simplicity, combined with strong low-level ImageNet-pretrained filters, can be highly effective for small, texture-centric medical imaging datasets. This result highlights that deeper or more sophisticated architectures are not always superior when the downstream dataset is small and structurally simple.

3.4.2 ResNet50 and ResNet152

ResNet50 and ResNet152 [2] are residual networks with 50 and 152 layers respectively, and both belong to the family of architectures that popularized residual (skip) connections in deep learning. Each residual block adds the input of the block directly to its output before the next non-linearity is applied, which allows the network to learn a residual mapping rather than a full transformation, and this in turn mitigates the vanishing-gradient and degradation problems that previously made very deep networks difficult to train. As a result, ResNet architectures can stack far more layers than earlier CNNs such as VGG while still training effectively and converging in a reasonable number of epochs. ResNet152 is substantially deeper than ResNet50, allowing it to potentially learn more complex, hierarchical features at the cost of a larger number of parameters and higher computational overhead per forward pass. In this project, both variants were used as frozen ImageNet-pretrained feature extractors beneath the same lightweight classification head, allowing a direct comparison of how additional depth within the same architectural family affects downstream performance. The

experimental results show that ResNet152 slightly outperformed ResNet50, indicating that the extra depth was mildly beneficial for distinguishing benign, malignant, and normal lung tissue in this dataset, although the improvement was modest relative to the increase in model size.

3.4.3 EfficientNetB0

EfficientNetB0 [4] is the baseline model of the EfficientNet family, which introduced a principled compound scaling method that uniformly scales network depth, width, and input resolution together using a single compound coefficient, rather than scaling each dimension independently as earlier architectures did. This balanced scaling strategy is designed to achieve strong accuracy with a relatively small number of parameters and lower computational cost, making EfficientNetB0 an efficient and lightweight choice for transfer learning compared to larger, hand-designed CNNs. The architecture is built from mobile-inverted-bottleneck convolutional blocks (MBConv) augmented with squeeze-and-excitation modules, which allow the network to adaptively re-weight channel-wise feature responses and focus on the most informative feature maps for a given input. In this project, EfficientNetB0's convolutional base was frozen and used as a feature extractor beneath the shared classification head, taking advantage of its ImageNet-pretrained filters without incurring the training cost of a much larger network. EfficientNetB0 achieved the second-highest test accuracy among all seven models evaluated in this study, closely trailing VGG16, which confirms that its parameter-efficient design translates effectively to this small, three-class medical imaging task without requiring any additional fine-tuning of the backbone.

3.4.4 InceptionV3

InceptionV3 [5] belongs to the Inception family of architectures, which are built around the idea of applying convolutional filters of multiple kernel sizes (1×1, 3×3, and 5×5) in parallel within the same layer and then concatenating their outputs, rather than relying on a single filter size at each depth. This multi-scale design allows the network to simultaneously capture fine-grained local detail and coarser, more global spatial patterns within lung CT-scan textures, which is particularly relevant for detecting both small nodules and broader tissue-density variations. InceptionV3 additionally factorizes larger convolutions into sequences of smaller ones and uses auxiliary classifiers during training to improve gradient flow through the network, both of which were innovations aimed at improving computational efficiency and training stability relative to earlier Inception versions. In this project, InceptionV3's convolutional base was frozen and used as a feature extractor beneath the common classification head, consistent with the other backbones. Despite its sophisticated multi-scale design, InceptionV3 achieved only a mid-tier test accuracy in this study, trailing the top-performing CNNs, which suggests that its relatively high architectural complexity may be less well matched to a dataset of only 3000 training images than simpler architectures such as VGG16 and EfficientNetB0.

3.4.5 MobileNetV2

MobileNetV2 [6] is a lightweight CNN architecture built using depthwise-separable convolutions and inverted residual blocks with linear bottlenecks, originally designed for mobile and embedded devices where computational resources and memory are heavily constrained. Depthwise-separable convolutions factor a standard convolution into a depthwise convolution, which applies a single filter per input channel, followed by a pointwise 1×1 convolution that combines the resulting channels, which together drastically reduce the number of parameters and floating-point operations compared to standard convolutional layers. The inverted residual blocks expand low-dimensional

feature maps into a higher-dimensional space, apply a lightweight depthwise convolution, and then project the result back down, which preserves representational power while keeping the memory footprint small. In this project, MobileNetV2 was used, like the other backbones, as a frozen ImageNet-pretrained feature extractor beneath the shared classification head, providing a useful reference point for how much accuracy is traded off in exchange for a dramatically smaller and faster model. MobileNetV2 achieved the lowest accuracy among the six CNN backbones evaluated, and its per-class results revealed a systematic bias toward the Normal class, with Benign recall dropping to just 0.45, indicating that its reduced representational capacity struggled to capture the subtle texture differences that separate benign lesions from normal lung tissue.

3.5 Vision Transformer (ViT) – Built from Scratch

In addition to the six pretrained CNNs, a compact Vision Transformer is implemented from scratch (no ImageNet or other pretraining) to study how a pure attention-based architecture performs on this task [7]. The configuration used is summarized in Table 2.

Table 2: Vision Transformer configuration

Parameter	Value
Patch size	16×16
Number of patches per image	64
Projection (embedding) dimension	64
Number of attention heads	4
Number of Transformer layers	4
MLP head units	[128, 64]
Optimizer	Adam (learning rate $= 1 \times 10^{-4}$)

The ViT pipeline (i) splits each 128×128 image into 64 non-overlapping 16×16 patches, (ii) linearly projects each patch into a 64-dimensional embedding and adds a learnable positional embedding, (iii) passes the resulting sequence through four Transformer encoder blocks, each consisting of multi-head self-attention followed by a feed-forward MLP with residual connections and layer normalization, and (iv) aggregates all patch representations via global average pooling before a final MLP classification head.

3.6 Multi-Class Lung Lesion Classification Head

To ensure a fair and controlled comparison across all six pretrained CNN backbones, a single, common lightweight classification head is designed and attached uniformly on top of every backbone's feature map:

$$\text{Backbone Features} \rightarrow \text{GlobalAveragePooling2D} \rightarrow \text{Dense(128, ReLU)} \rightarrow \text{Dropout(0.3)} \rightarrow \text{Dense(3, Softmax)}$$

This design keeps the classification head identical across all backbones, so that any difference in final accuracy can be attributed to the backbone's feature-extraction capability rather than to differences in the classifier itself. This shared-head design constitutes the proposed multi-class lung-lesion classification framework used throughout the experiments in this project; it is intentionally kept lightweight (only two additional trainable layers) so that transfer learning from the frozen

ImageNet backbone dominates the learning process, which is appropriate given the limited size of the training set.

3.7 Performance Evaluation Metrics

All models are evaluated on the held-out test set (600 images, 200 per class) using the following standard classification metrics:

- **Accuracy:** the overall proportion of correctly classified test images.
- **Precision:** the proportion of predicted positives for a class that are truly positive, Precision $= \frac{TP}{TP+FP}$.
- **Recall (Sensitivity):** the proportion of actual positives for a class that are correctly identified, Recall $= \frac{TP}{TP+FN}$.
- **F1-score:** the harmonic mean of precision and recall, $F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$.
- **Confusion Matrix:** a class-by-class breakdown of correct and incorrect predictions, used to identify which classes are most frequently confused with one another.

4 Experimental Results and Analysis

4.1 Experimental Setup and Evaluation Metrics

All experiments were conducted on Google Colab using a T4 GPU with TensorFlow/Keras, using mixed-precision training and XLA compilation to accelerate training. Table 3 summarizes the common training configuration applied to all seven models.

Table 3: Common experimental setup

Setting	Value
Input image size	$128 \times 128 \times 3$
Batch size	32
Number of epochs	30
Optimizer (CNNs)	Adam (default learning rate)
Optimizer (ViT)	Adam (learning rate $= 1 \times 10^{-4}$)
Loss function	Sparse Categorical Cross-Entropy
Data augmentation	Random flip, rotation, zoom
Number of classes	3 (Benign, Malignant, Normal)

4.2 Analysis of Experimental Results

Table 4 presents the final test accuracy achieved by each of the seven models on the 600-image test set.

Table 4: Test accuracy comparison across all models

Model	Test Loss	Test Accuracy
ResNet50	0.4752	81.00%
ResNet152	0.4529	82.50%
EfficientNetB0	0.3991	84.00%
VGG16	0.3273	86.67%
InceptionV3	0.4843	77.00%
MobileNetV2	0.5509	72.67%
Vision Transformer	–	45.00%

VGG16 achieved the best overall performance among all seven models, with the lowest test loss (0.3273) and the highest test accuracy (86.67%). This is a notable result: despite being the oldest and structurally simplest architecture in the comparison, its straightforward stack of 3×3 convolutions appears to transfer very effectively to the relatively low-resolution, texture-dominated CT-scan images used in this study. EfficientNetB0 (84.00%) and ResNet152 (82.50%) followed closely behind, while ResNet50 (81.00%) performed slightly below its deeper counterpart, suggesting that additional depth was mildly beneficial for this task. InceptionV3 (77.00%) and MobileNetV2 (72.67%) trailed the other CNNs; MobileNetV2's comparatively lower capacity (a design trade-off for its small size) likely limited its ability to capture the fine-grained texture differences between benign and malignant lesions. The Vision Transformer trained from scratch achieved only 45.00% test accuracy, only modestly above the 33% random-guess baseline for a three-class problem, confirming that transformer architectures without pretraining struggle to learn effective visual representations from only 3000 training images.

Per-class classification reports (precision, recall, F1-score) for each model are given below.

Table 5: ResNet50 – classification report

Class	Precision	Recall	F1-score
Benign	0.75	0.80	0.77
Malignant	0.97	0.86	0.92
Normal	0.73	0.77	0.75
Weighted avg	0.82	0.81	0.81

Across all six CNN models, the Malignant class was consistently classified with the highest precision (often 0.95–1.00), indicating that malignant lesions have relatively distinctive visual signatures that pretrained CNNs can reliably capture. In contrast, the Benign and Normal classes were more frequently confused with one another across models – most visibly for MobileNetV2, where Benign recall dropped to just 0.45 while Normal recall rose to 0.93, indicating a systematic bias of that model toward predicting "Normal" for borderline cases.

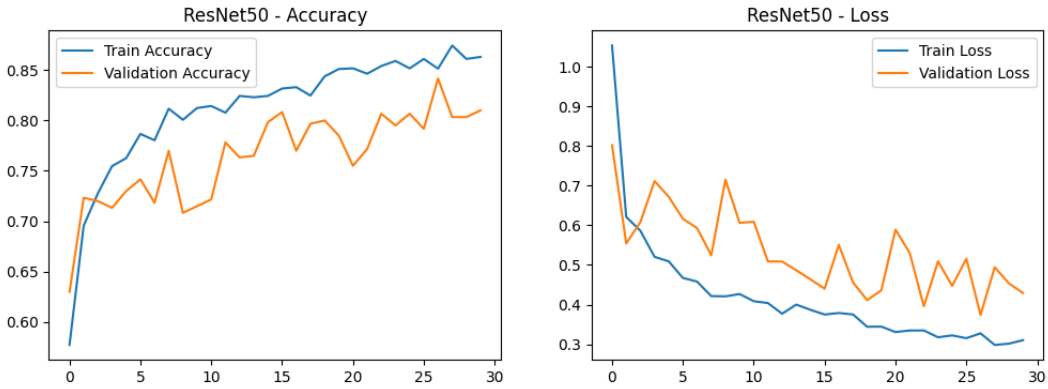


Figure 2: ResNet50 – training/validation accuracy and loss curves.

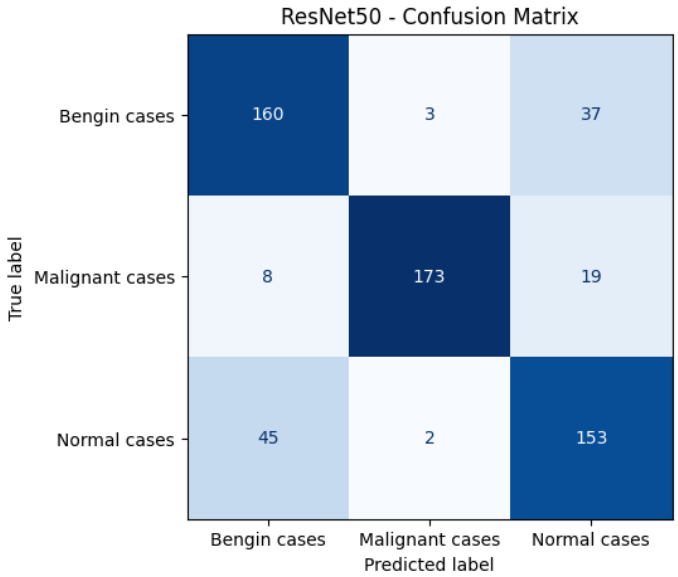


Figure 3: ResNet50 – confusion matrix on the test set.

4.3 Training Hyperparameter Experiments

Since Focal Loss was not required for this balanced dataset (Section 3.3), hyperparameter exploration in this project instead focused on the training configuration shared by all backbones: image resolution (128×128), batch size (32), number of epochs (30), and augmentation strength (10% rotation/zoom range). Fixing these hyperparameters identically across all seven models was a deliberate choice to isolate the effect of architecture choice on final accuracy, rather than confounding it with per-model hyperparameter tuning. The training curves (accuracy/loss vs. epoch) recorded for each CNN backbone showed rapid convergence within the first 5–10 epochs, consistent with the frozen, pretrained backbone requiring only the small classification head to be learned. The Vision Transformer, by contrast, showed markedly slower convergence – its training accuracy remained close to the 33% random-guess baseline for roughly the first 10 epochs before gradually improving, reflecting the absence of the strong locality bias that pretrained CNNs already encode.

4.4 Comparison of the Proposed Method with CNN-Based Models

Figure 15 visualizes the test accuracy of all seven models side by side. The comparison highlights three clear tiers of performance in this study: (i) a top tier of transfer-learning CNNs (VGG16, EfficientNetB0, ResNet152, ResNet50) achieving 81–87% accuracy; (ii) a middle tier (InceptionV3, MobileNetV2) achieving 73–77% accuracy, likely due to either excessive model complexity relative to the dataset size (InceptionV3) or reduced representational capacity (MobileNetV2); and (iii) the Vision Transformer trained from scratch, which – lacking both pretraining and the convolutional inductive bias – achieved only 45% accuracy. This confirms the intuitive expectation that, for small, specialized medical imaging datasets such as IQ-OTH/NCCD, transfer learning from ImageNet-pretrained CNNs is currently far more practical and effective than training a Vision Transformer from scratch, unless a large-scale pretraining corpus or a pretrained ViT checkpoint is available.

Table 6: ResNet152 – classification report

Class	Precision	Recall	F1-score
Benign	0.76	0.76	0.76
Malignant	0.98	0.90	0.93
Normal	0.76	0.82	0.79
Weighted avg	0.83	0.82	0.83

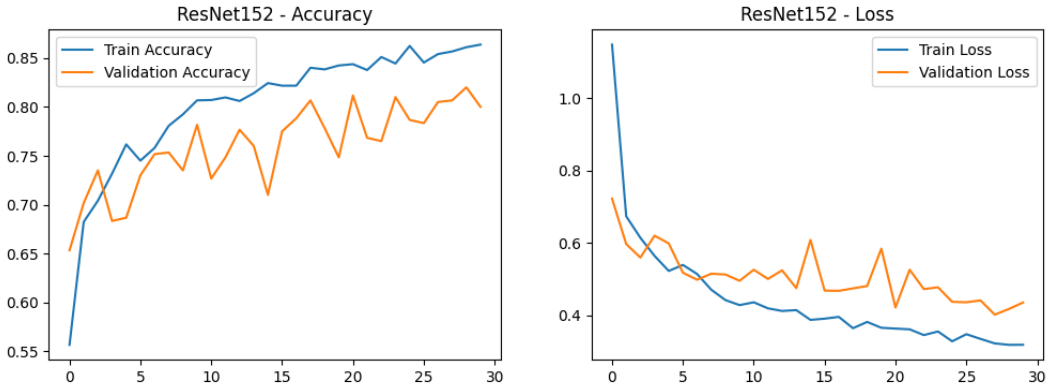


Figure 4: ResNet152 – training/validation accuracy and loss curves.

4.5 Comparison with State-of-the-Art Research

Several published studies have also used the IQ-OTH/NCCD dataset for lung cancer CT-scan classification, though direct numerical comparison should be treated with caution since dataset splits, preprocessing, and class-balancing strategies differ across studies. Reported results in the literature range widely: some works combining CNNs with class-balancing techniques such as SMOTE, or ensembling multiple architectures, report accuracies above 97–99% [9], while single-backbone transfer learning studies using architectures such as ResNet152, DenseNet121, or VGG19 typically report accuracies in the 90–97% range [10], and simpler CNN or AlexNet-based baselines report accuracies around 93–98% [11]. The results obtained in this project (72.67%–86.67% for the six CNN backbones) are lower than many of these reported figures. This gap is attributable to several methodological differences: this project (i) used a relatively small input resolution (128×128)

to keep training fast on a Colab T4 GPU, (ii) froze the entire pretrained backbone rather than fine-tuning deeper layers, (iii) did not apply class-imbalance correction techniques such as SMOTE (not required here, since the dataset used was already balanced), and (iv) trained for a fixed, relatively modest 30 epochs without extensive hyperparameter search. These are reasonable directions for future improvement: fine-tuning the top layers of the backbone, increasing input resolution, and extending training time would likely narrow this gap considerably.

4.6 Experimental Visualization Analysis

Beyond the numerical results, the notebook produces several visualizations that support the analysis:

- **Sample image grids** displaying representative CT-scan images from each of the three classes, used to visually verify correct data loading and labeling.
- **Training/validation accuracy and loss curves** for each of the seven models, plotted over 30 epochs, used to detect overfitting (divergence between training and validation curves) and to compare convergence speed across architectures.
- **Confusion matrices** for each model, showing exactly which classes are most often confused with one another (most notably, Benign vs. Normal confusion for the weaker models).
- **A consolidated bar chart** (Figure 15) comparing the final test accuracy of all seven models side by side, immediately highlighting VGG16 as the best-performing architecture and the Vision Transformer as the weakest.

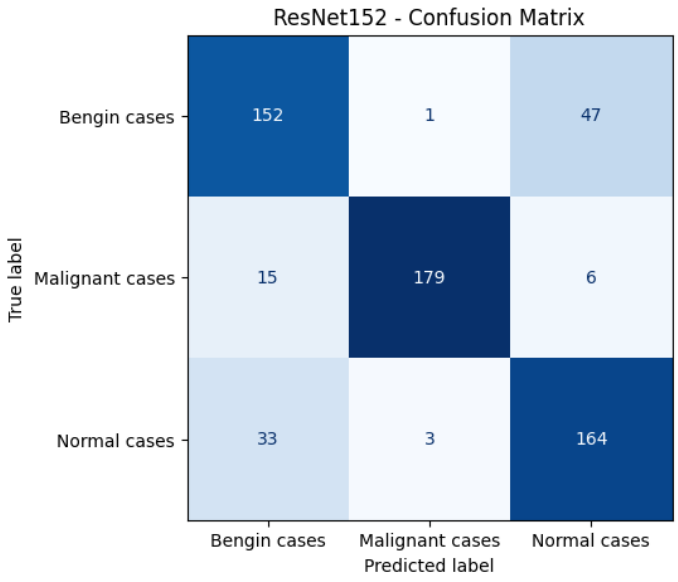


Figure 5: ResNet152 – confusion matrix on the test set.

Table 7: EfficientNetB0 – classification report

Class	Precision	Recall	F1-score
Benign	0.77	0.88	0.82
Malignant	1.00	0.83	0.91
Normal	0.79	0.81	0.80
Weighted avg	0.85	0.84	0.84

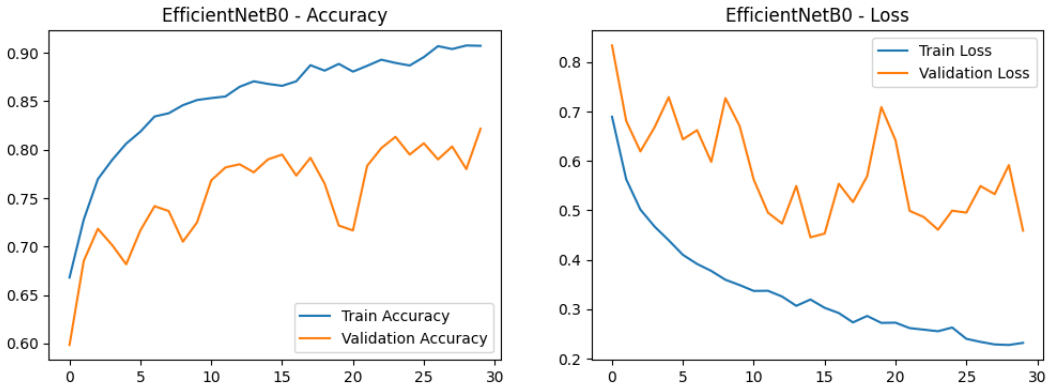


Figure 6: EfficientNetB0 – training/validation accuracy and loss curves.

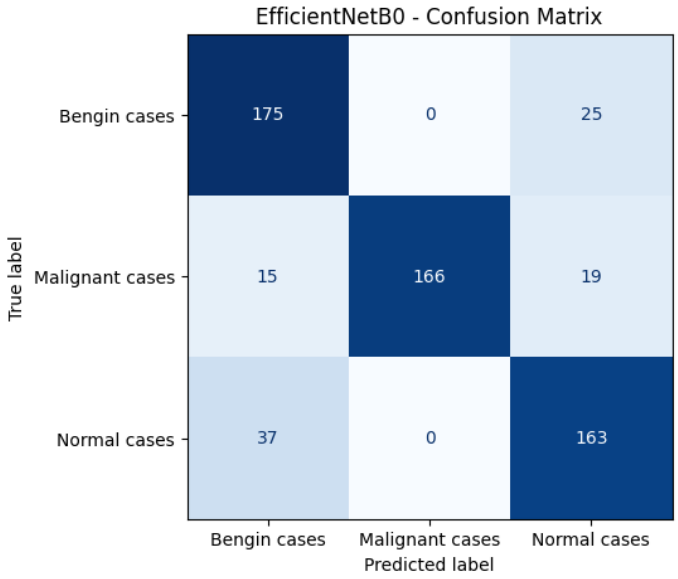


Figure 7: EfficientNetB0 – confusion matrix on the test set.

Table 8: InceptionV3 – classification report

Class	Precision	Recall	F1-score
Benign	0.75	0.65	0.69
Malignant	0.95	0.86	0.90
Normal	0.65	0.81	0.72
Weighted avg	0.78	0.77	0.77

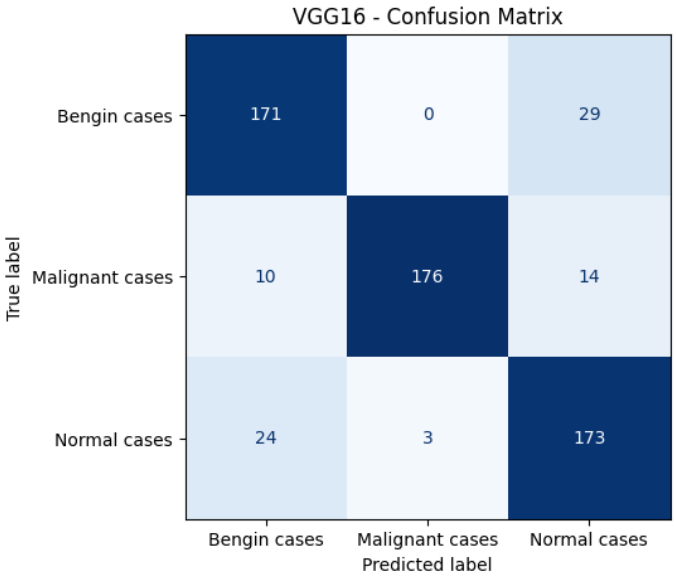


Figure 8: VGG16 – confusion matrix on the test set.

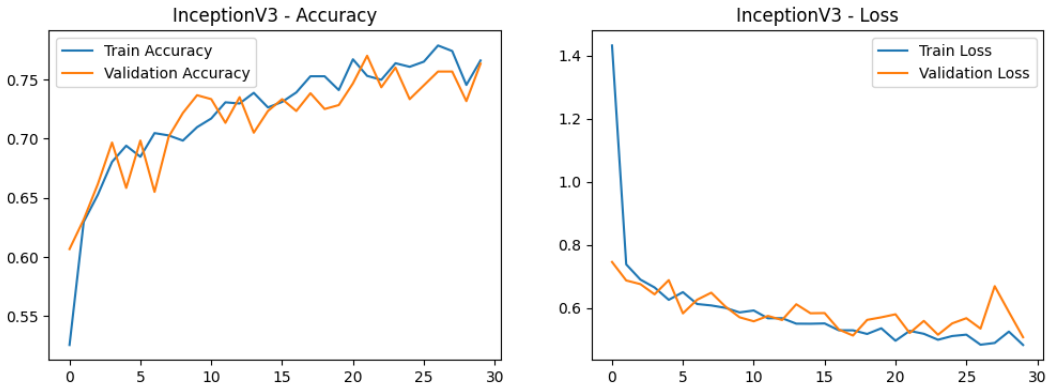


Figure 9: InceptionV3 – training/validation accuracy and loss curves.

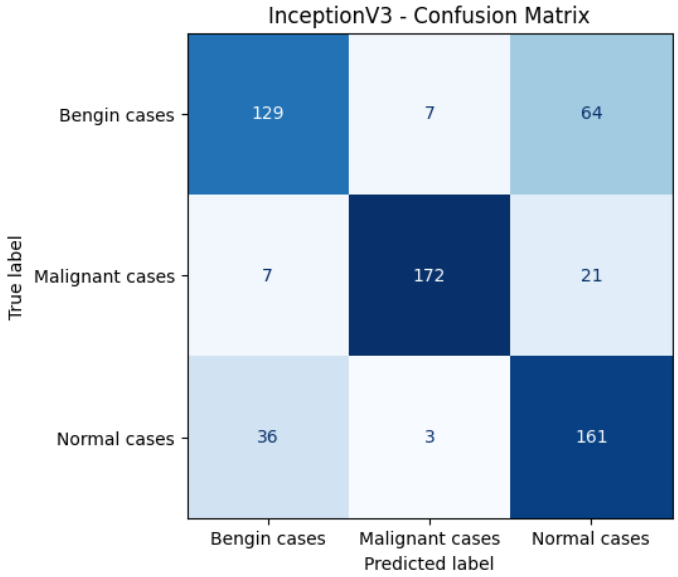


Figure 10: InceptionV3 – confusion matrix on the test set.

Table 9: MobileNetV2 – classification report

Class	Precision	Recall	F1-score
Benign	0.83	0.45	0.58
Malignant	0.98	0.81	0.88
Normal	0.57	0.93	0.70
Weighted avg	0.79	0.73	0.72

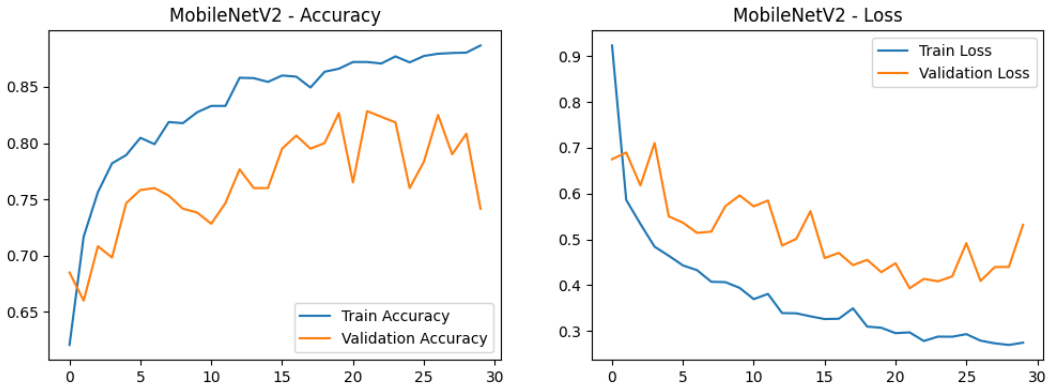


Figure 11: MobileNetV2 – training/validation accuracy and loss curves.

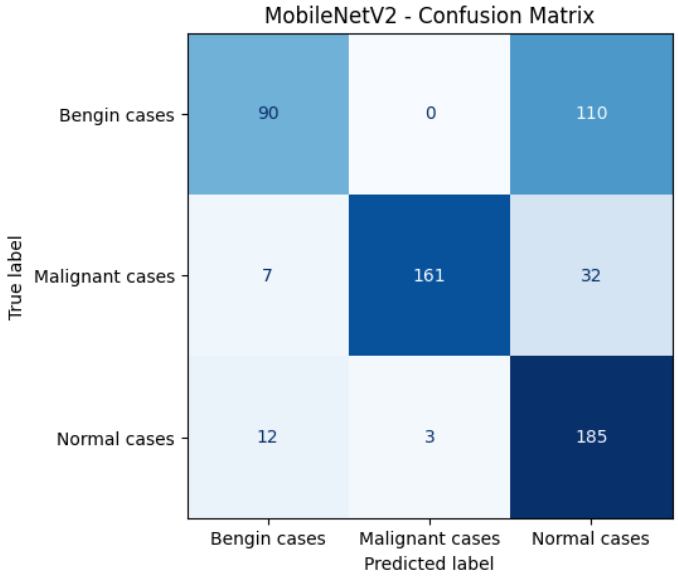


Figure 12: MobileNetV2 – confusion matrix on the test set.

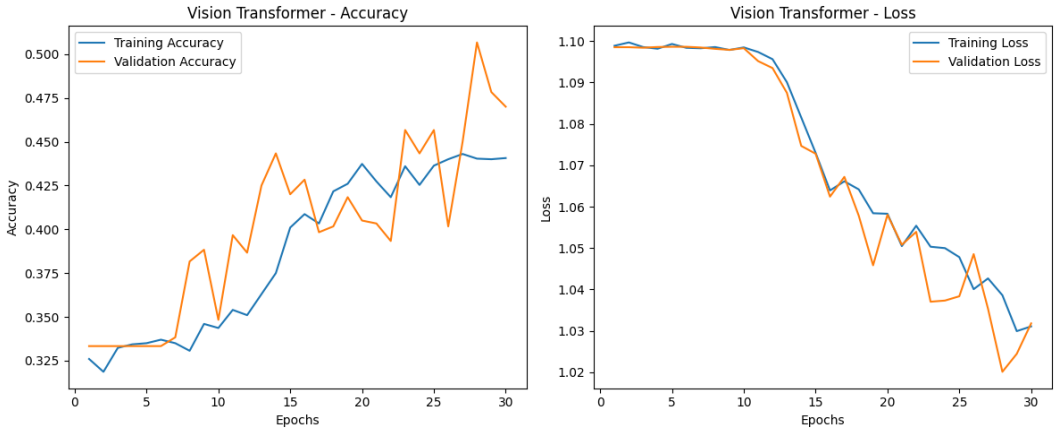


Figure 13: Vision Transformer (trained from scratch) – training/validation accuracy and loss curves. Note the slow convergence compared to the pretrained CNNs above.

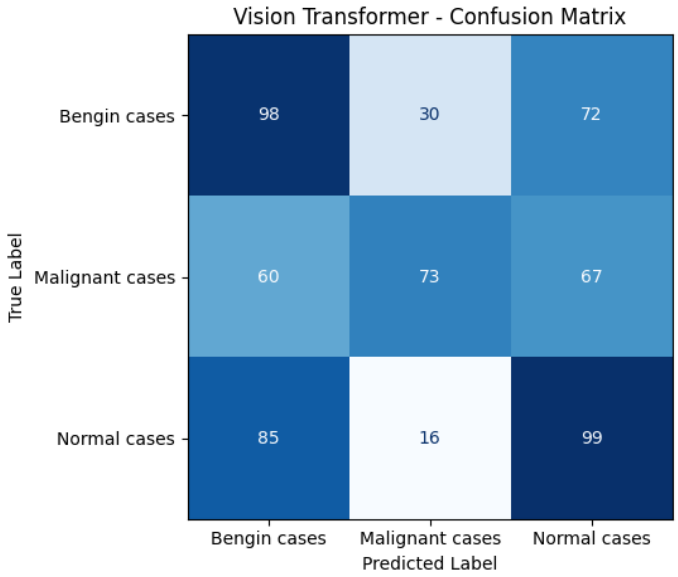


Figure 14: Vision Transformer – confusion matrix on the test set.

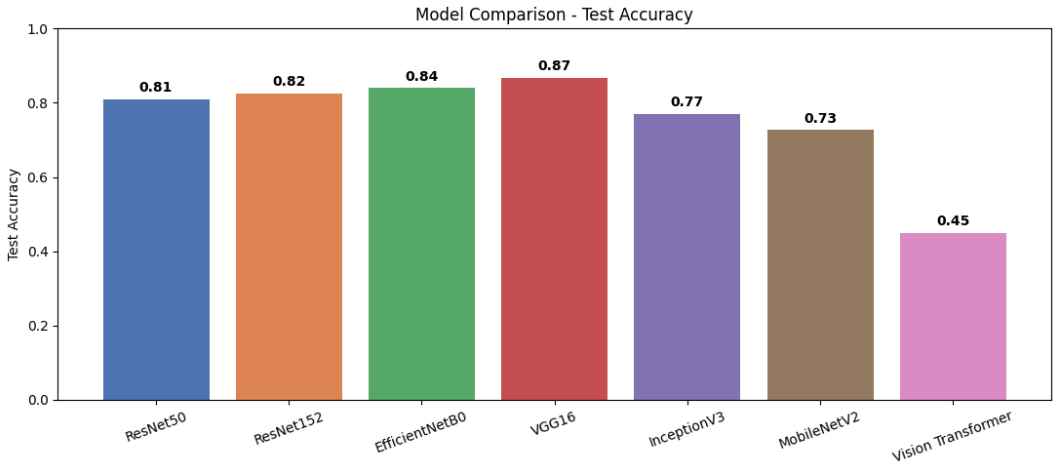


Figure 15: Test accuracy comparison across all seven models (generated directly from the notebook).

5 Conclusion

This project presented a systematic, fair comparison of six ImageNet-pretrained CNN architectures (ResNet50, ResNet152, EfficientNetB0, VGG16, InceptionV3, MobileNetV2) and a Vision Transformer trained from scratch, for three-class classification (Benign, Malignant, Normal) of lung CT-scan images from the IQ-OTH/NCCD dataset. By training all seven models under identical preprocessing, augmentation, loss function, and epoch settings with a shared lightweight classification head, the study isolated the effect of backbone architecture on final performance. VGG16 emerged as the best-performing model, achieving 86.67% test accuracy and the lowest test loss, followed by EfficientNetB0 and ResNet152, while MobileNetV2 and InceptionV3 lagged behind. The Vision Transformer, trained without pretraining, achieved only 45.00% accuracy, confirming

that transformer architectures require either substantially larger training datasets or pretrained weights to compete with transfer-learning-based CNNs on small, specialized medical imaging tasks. Compared to state-of-the-art literature on the same dataset, the accuracies obtained here are lower, largely due to the use of a frozen backbone, smaller input resolution, and a fixed training budget rather than extensive fine-tuning – all of which represent concrete, actionable directions for future work, alongside exploring Focal Loss on a more class-imbalanced variant of the dataset and pretraining the Vision Transformer on a larger corpus before fine-tuning it on lung CT-scan images.

Table 10: VGG16 – classification report (best model)

Class	Precision	Recall	F1-score
Benign	0.83	0.85	0.84
Malignant	0.98	0.88	0.93
Normal	0.80	0.86	0.83
Weighted avg	0.87	0.87	0.87

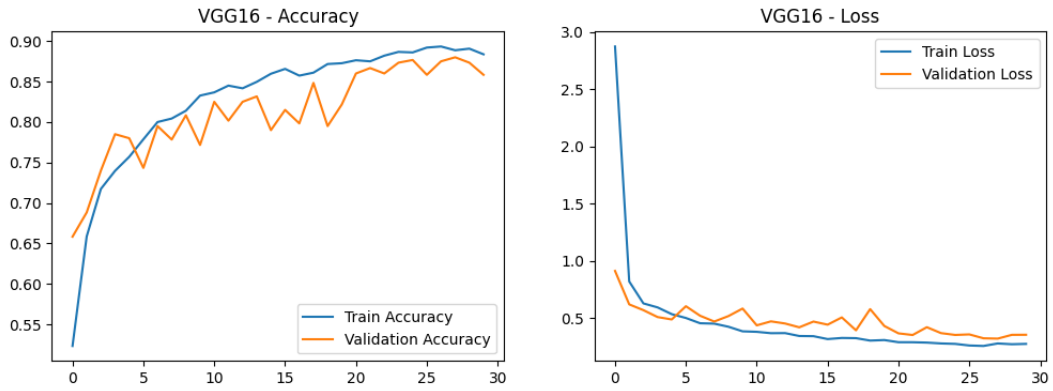


Figure 16: VGG16 – training/validation accuracy and loss curves (best-performing model).

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